Sprint 1: Flu Vaccination Predictor Project

By ALPHA GROUP
Assimagbe Albert Raphael & Saad Ali

Based on CDC H1N1 2009 Survey Data

Project Goals & Questions

- **Objectives:**
- - Identify who gets flu vaccines and why
- - Understand role of doctor recommendations
- - Use machine learning to predict vaccine behavior
- - Suggest smart ways to improve vaccination

- **Key Questions:**
- - Who avoids the vaccine?
- - Do doctor suggestions help?
- - Which behaviors matter most?

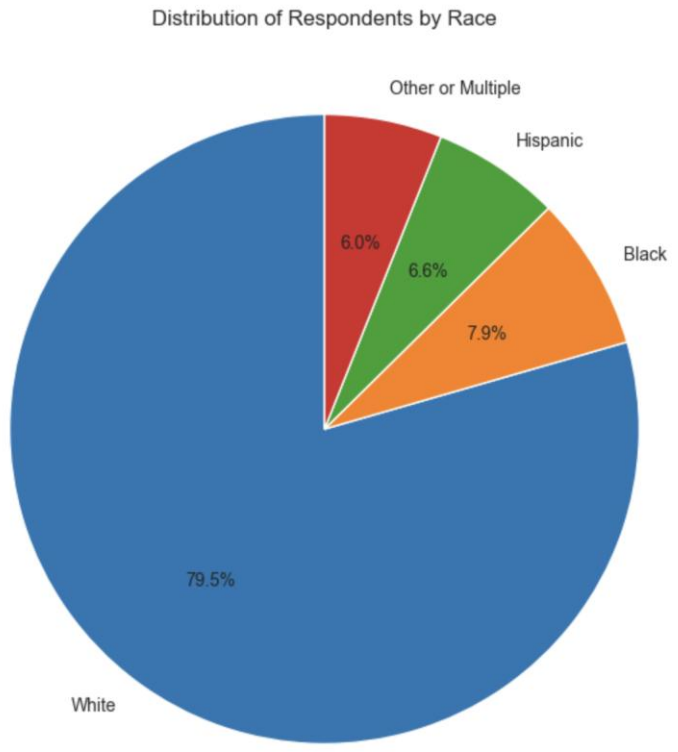
Dataset Summary

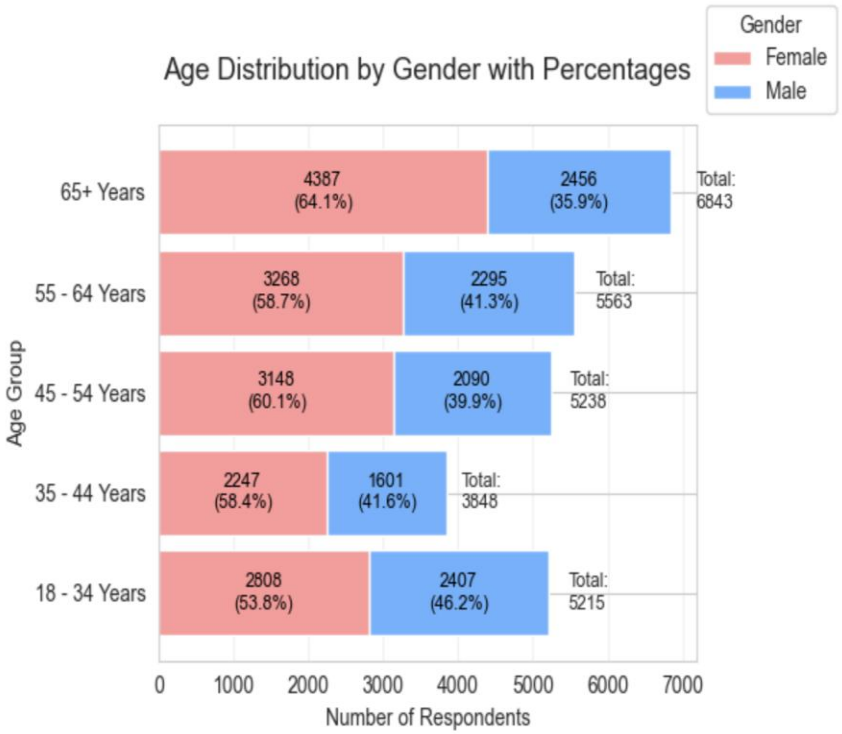
- Source: CDC 2009 H1N1 Survey
- Size: 26,707 people, 36 features

- **Data Includes:**
- - Demographics (age, education, race, gender, etc.)
- - Behaviors (handwashing, avoidance)
- - Health opinions and access

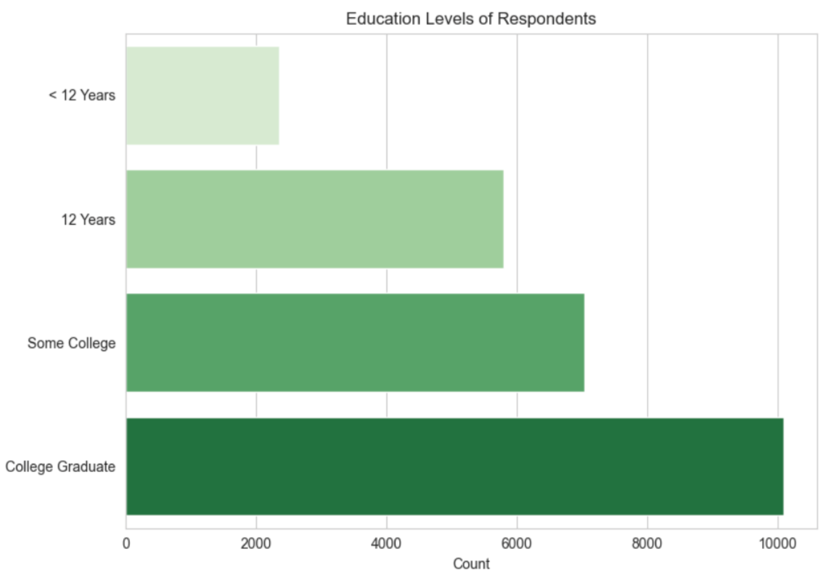
- **Challenge:** Missing data in some columns (up to 50%)

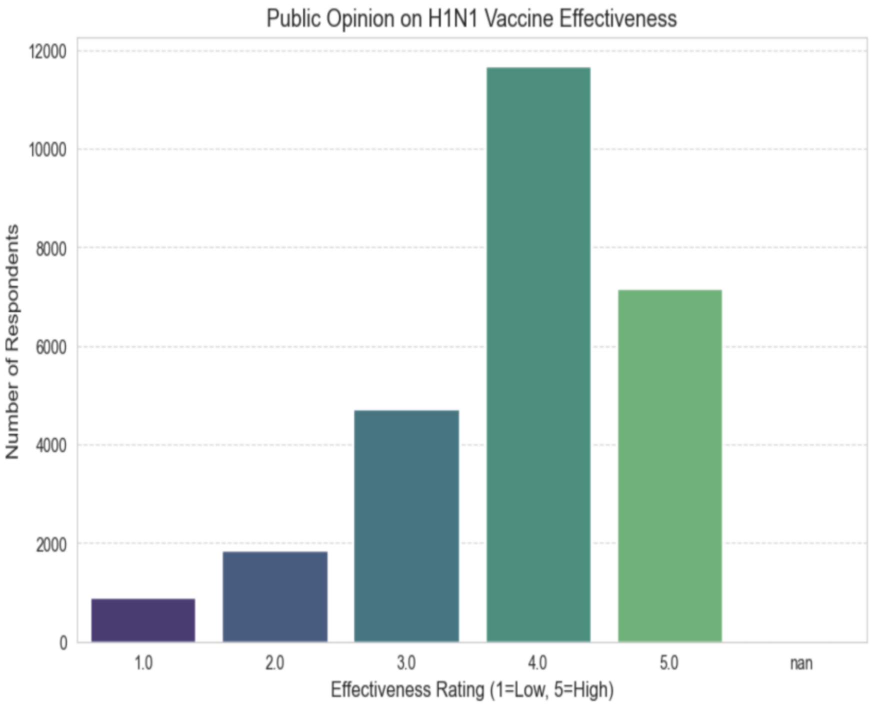
Visuals Of Race & Age Distributions





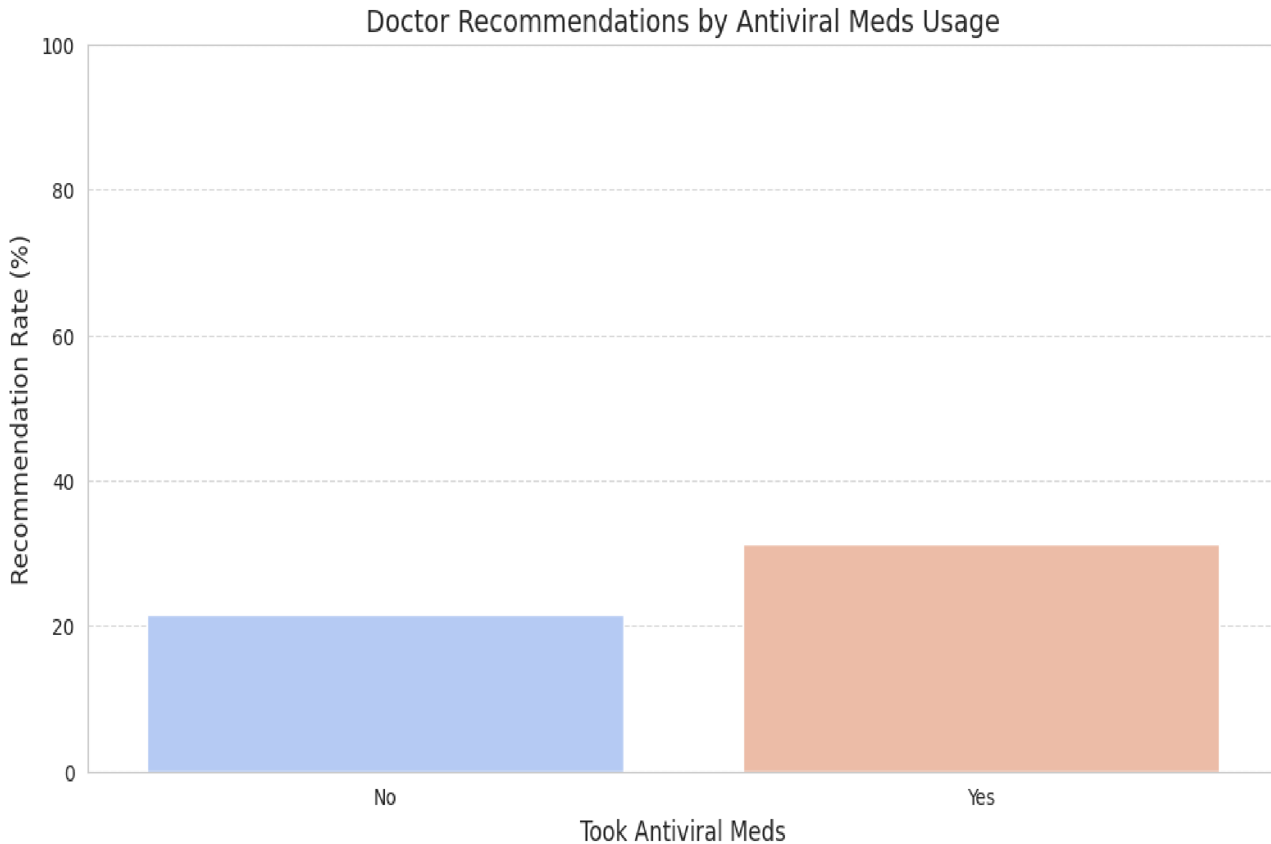
Visuals Of Educational Levels & Public Opinion





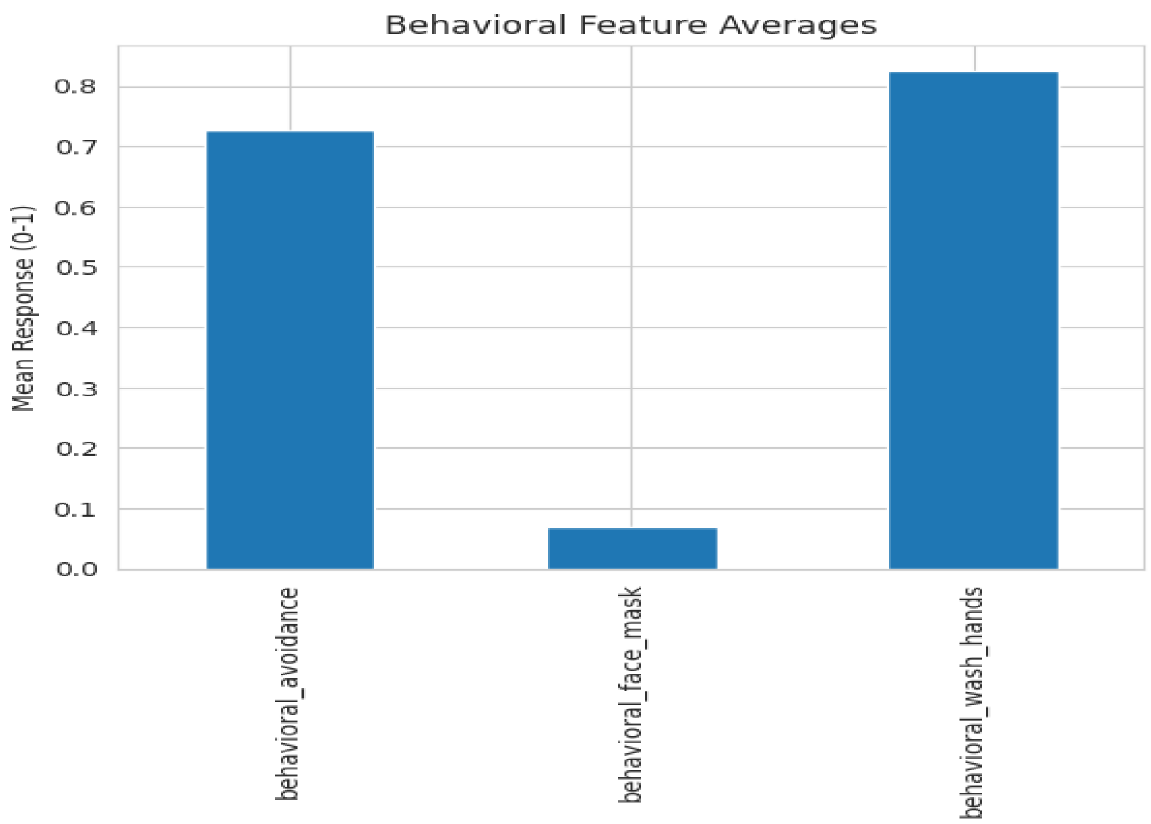
Key Findings & Visuals

1. Doctor Recommendations Influence Vaccination



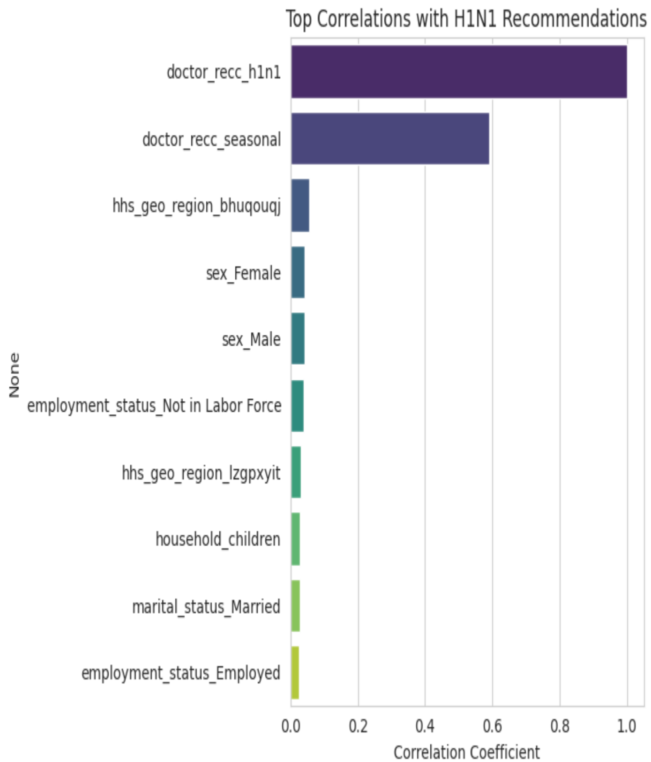
Key Of Findings

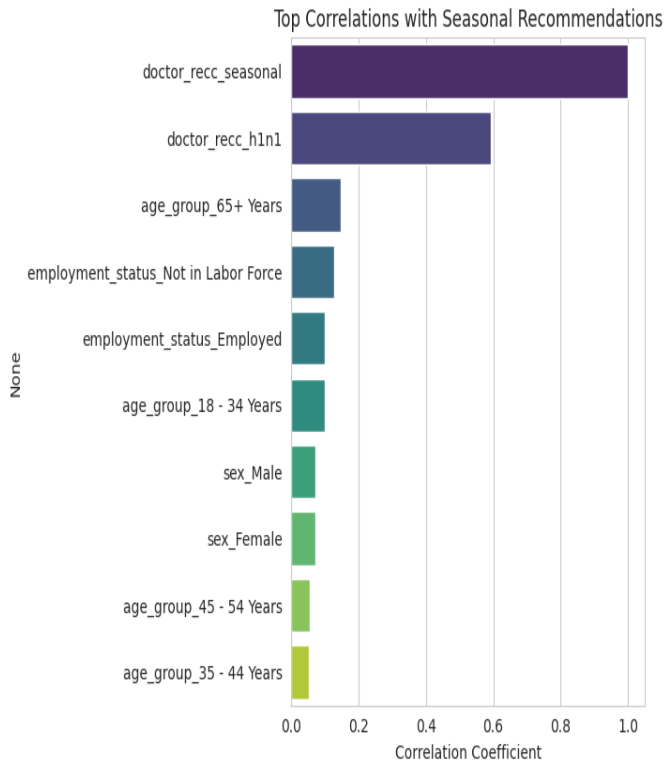
2. Concern about H1N1 Virus = more handwashing



Key Findings & Visuals

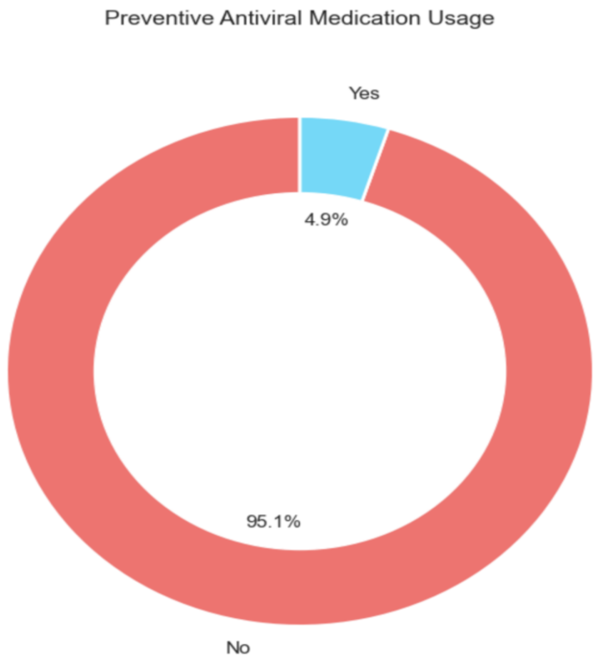
3. Older adults (65+) more likely to get seasonal flu shots

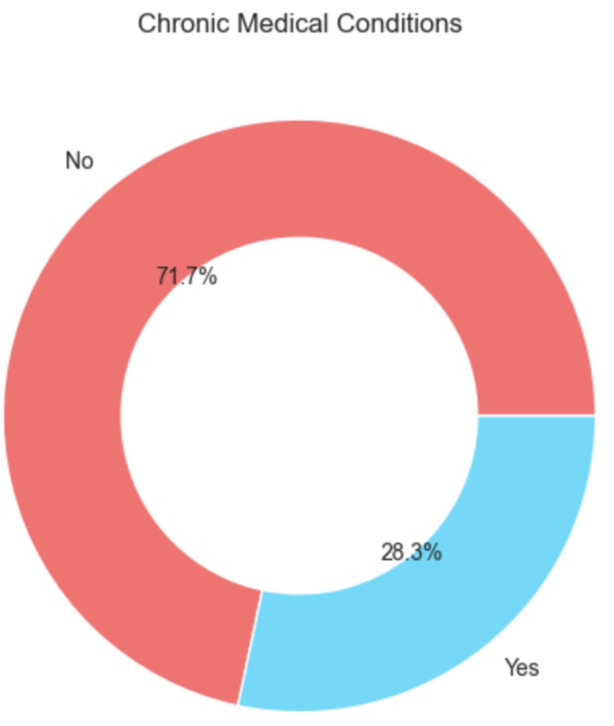




Key Findings & Visuals

4. 4.9% Persons takes Preventive Antiviral Medication while, 28.3% Persons has chronic medical conditions





Data Analysis Summary

- **Cleaned all missing data:**
- -Dropped high-missing columns (employment_occupation (50 %), employment_industry (49.9%), health_insurance (45.9%), etc.)
- **Important Analysis:**
- - Doctor recommendation (47.5%)
- - H1N1 Virus concern – face mark, wash hand and avoidance (57.2%)
- - Vaccine effectiveness belief
- - Preventive Antiviral Medical Usage (4.9%)
- - Chronic conditions (28.3%)

Recommendations

1. Train doctors to give strong, clear vaccine advice
2. Target under-vaccinated groups (50–64 year-olds)
3. Use SMS campaigns and infographics
4. Promote combined vaccine confidence messages

Final Thoughts

✅ Machine learning can help target interventions

✅ Beliefs and doctor input matter more than demographics

📢 Next step: Build prediction tool for clinics